Trouver le faux

Développement de modèles
Machine Learning pour détecter les billets de banque frauduleux



ONCFM

21 Mai 2026

Matilda Haraldsson

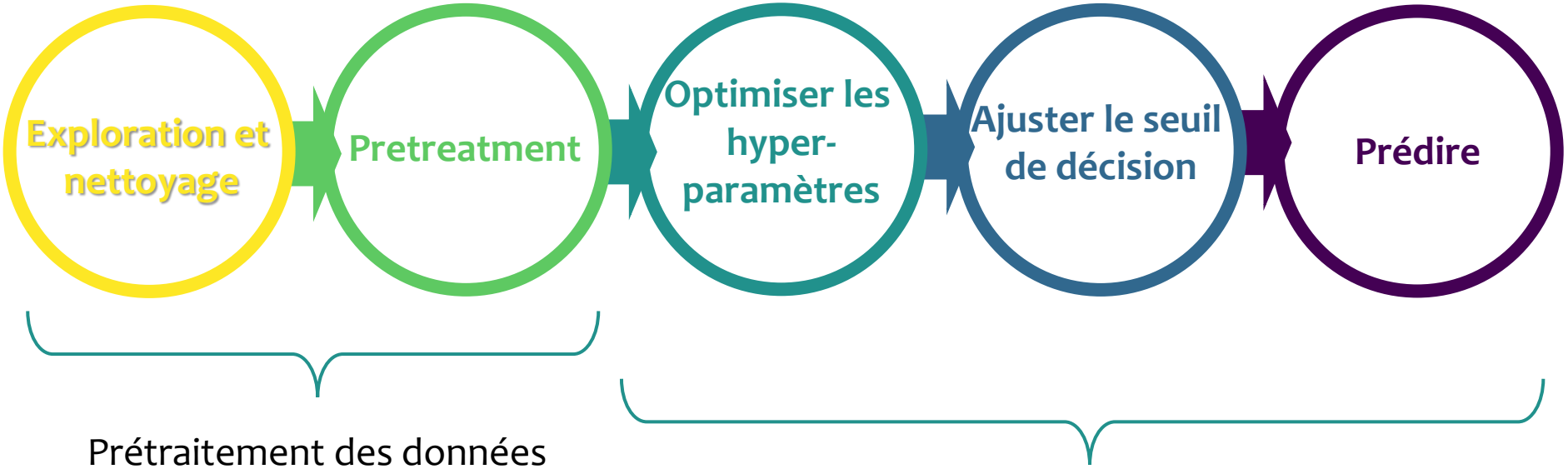
Objectives

- ONCFM: **Lutter** contre la **contrefacon** de billets en euros
- Mise en place des **méthodes d'identification** automatique des faux billets

➡ Développer un modèle **d'apprentissage automatique** (Machine Learning, ML) qui, à partir des caractéristiques géométriques sur le billet, **identifie les billets frauduleux**

- L'algorithes: Régression logistique, KNN, K-means, Random Forest

Flux de travail de Machine Learning



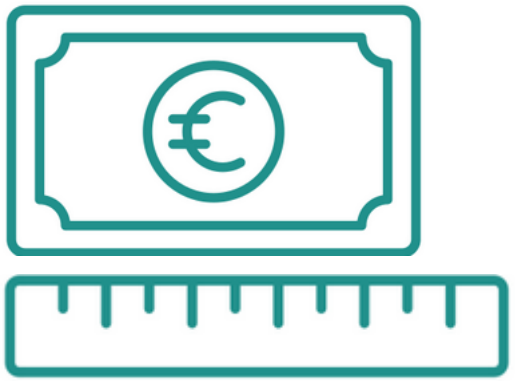
Nos données

Exploration et nettoyage

Données de **1500 billets de banque**

- **Six caractéristiques géométriques (variables prédictives)** :
  margin low / margin up / length
  height left / height right / diagonal (en mm)

- **Si authentique (variable cible)** ➡ is Fake (1000 vrais et 500 faux)



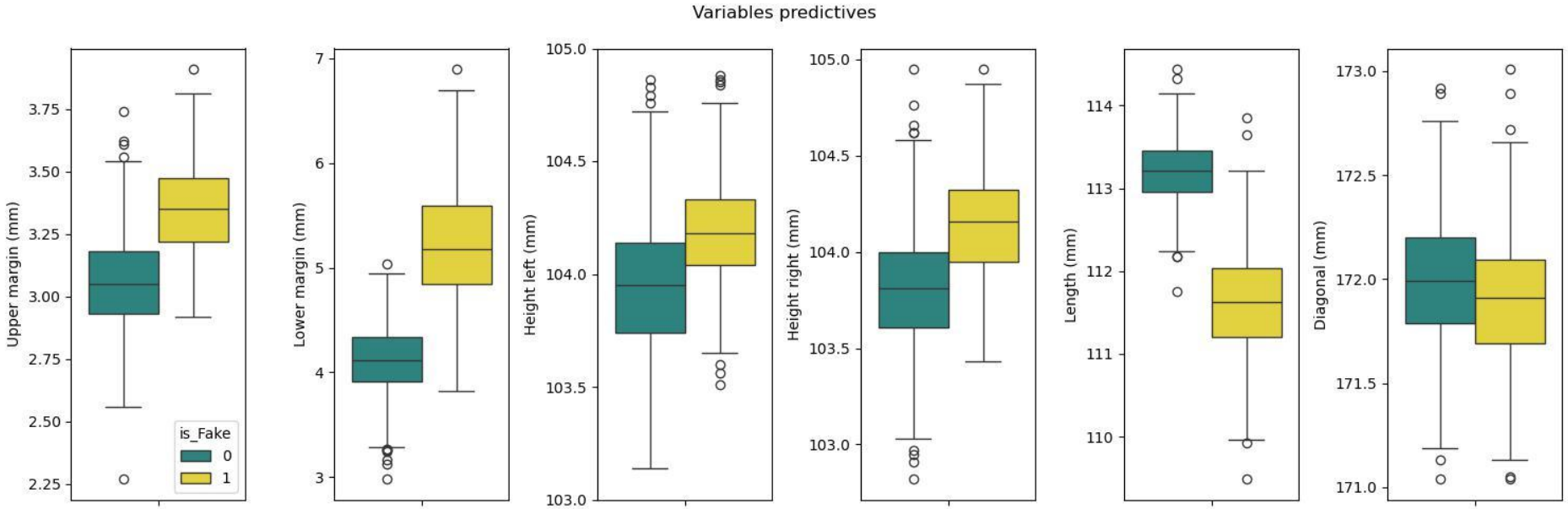


Nos données

Exploration et nettoyage

Données de **1500 billets de banque**

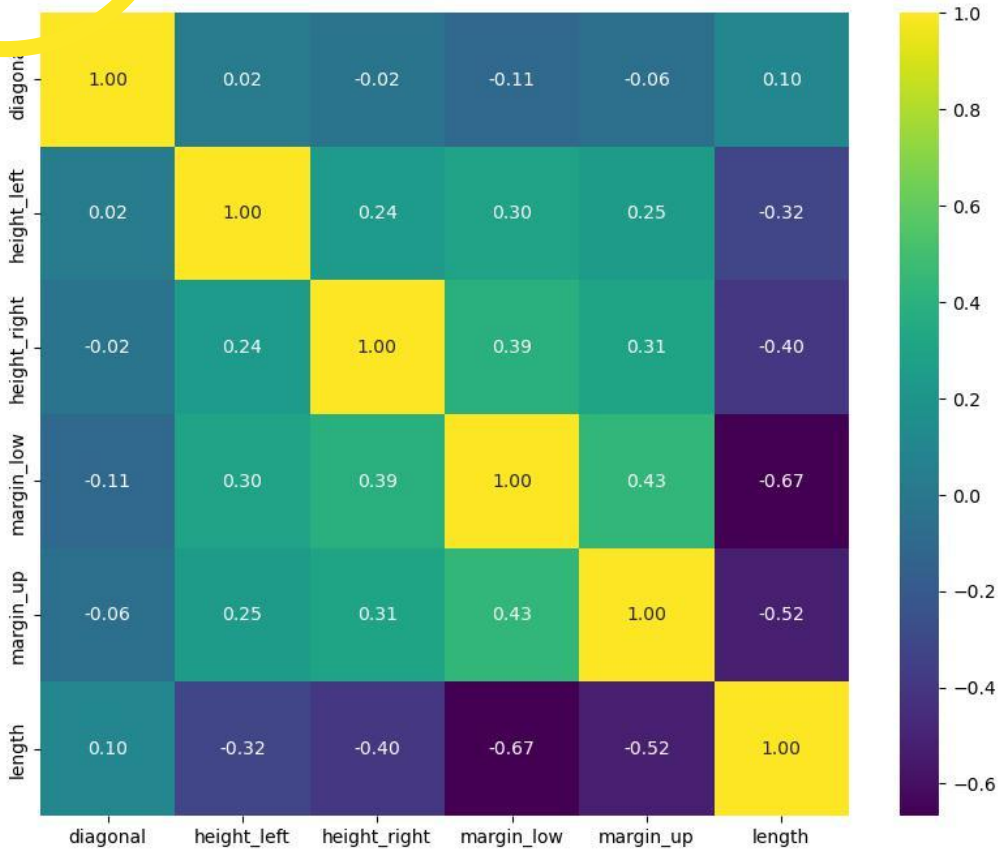
- **Six caractéristiques géométriques (variables prédictives)** :
  margin low / margin up / length
  height left / height right / diagonal (en mm)



Nos données

Exploration et nettoyage

- La « **margin low** » comporte 36 valeurs manquantes
- Cette variable est corrélée à la « **length** », et à la « **margin up** »



Imputation de données

Exploration et nettoyage

- La « margin low » comporte 36 valeurs manquantes
- Cette variable est corrélée à la
- Trois modèles de **régressions classique**

1) margin low ~ length

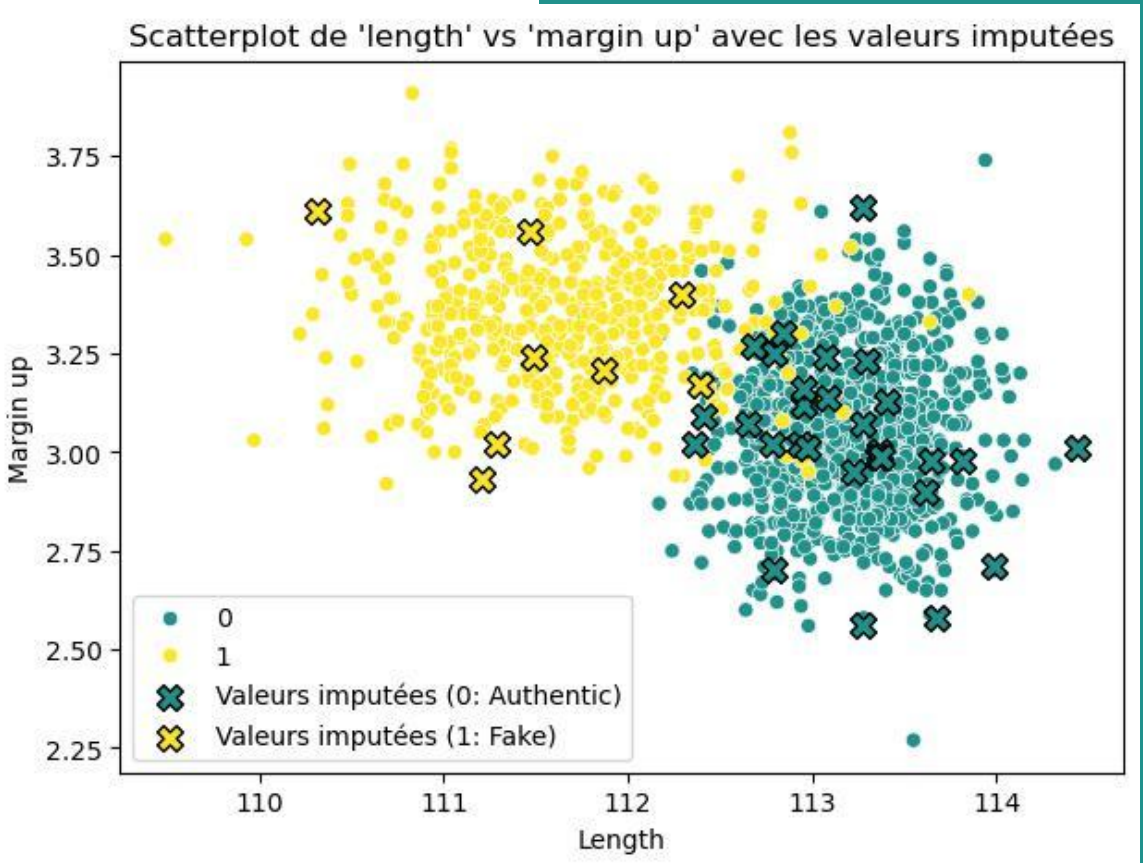
   R^2: 0.445, AIC: 2096

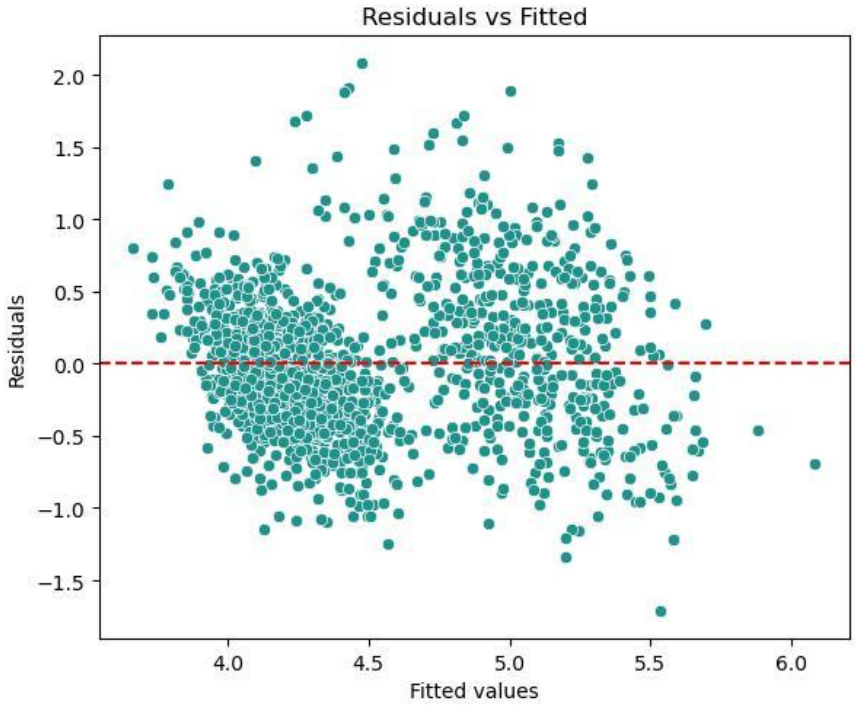
2) margin low ~ length + margin up

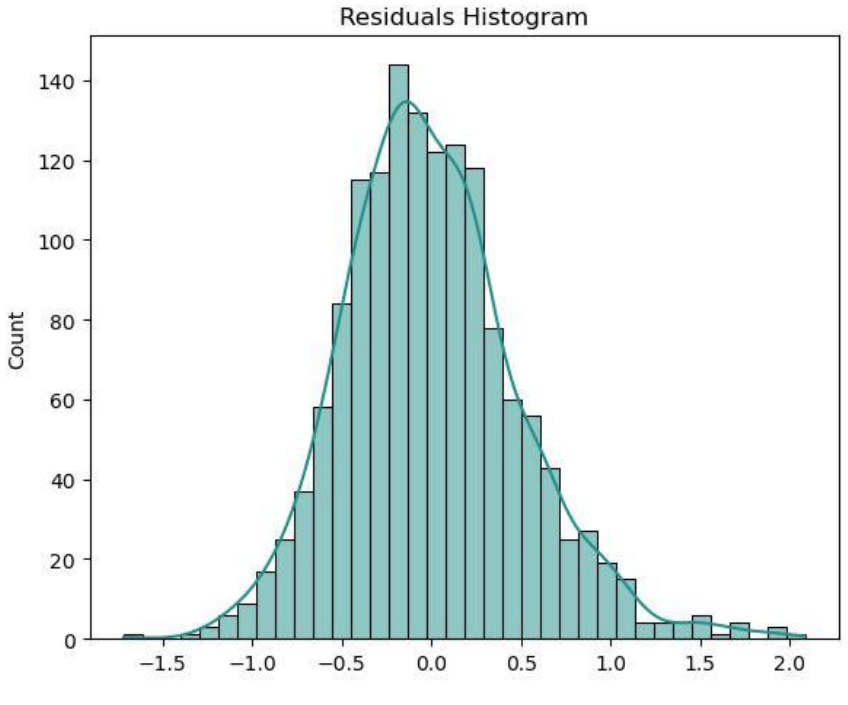
   R^2 : 0.454, AIC: 2072

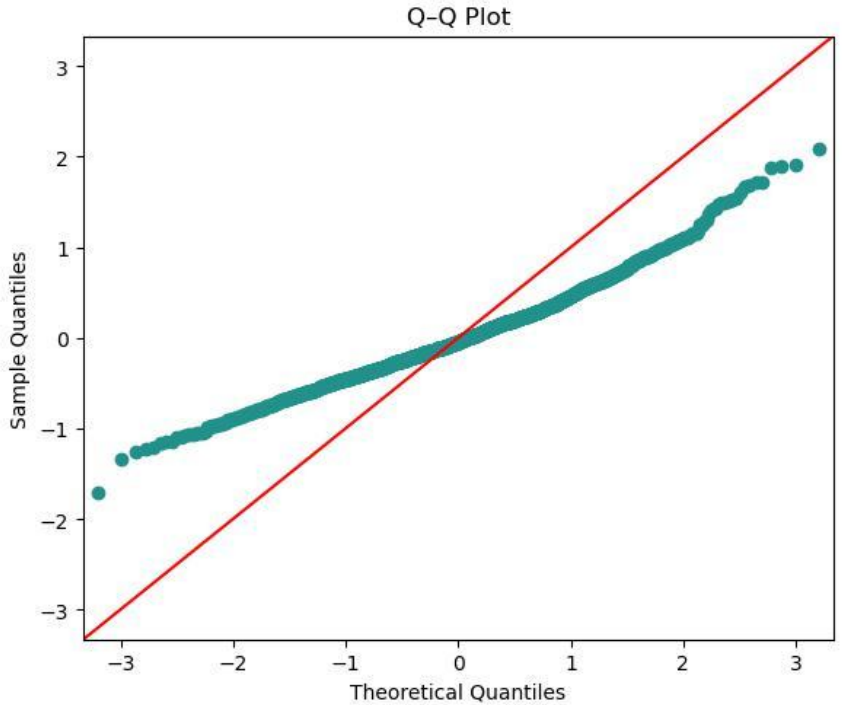
3) margin low ~ length + margin up + height right . …

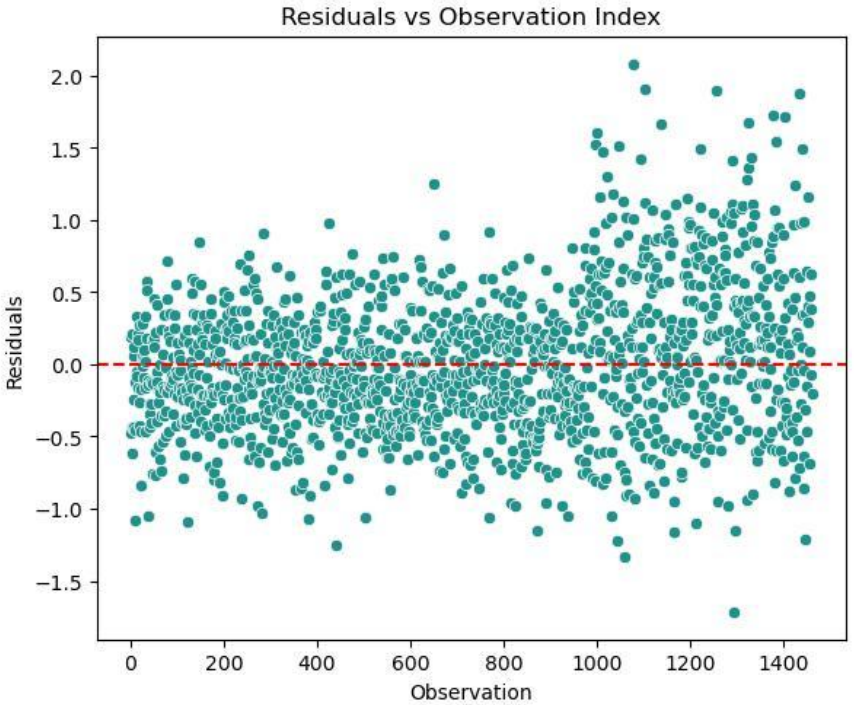
   R^2 : 0.477, AIC: 2015




Residuals vs Fitted
Residuals
Fitted values


Residuals Histogram
Count


Q–Q Plot
Sample Quantiles
Theoretical Quantiles


Residuals vs Observation Index
Residuals
Observation

Exploration et nettoyage

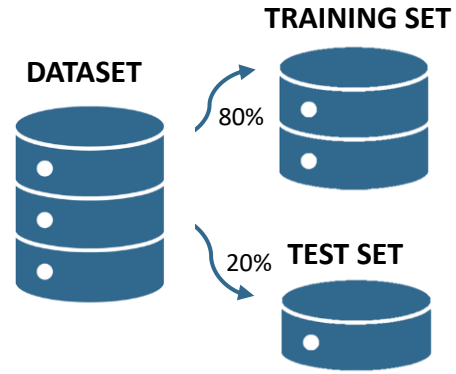
Pretreatment

- **Définir des variables:**

y : variables cible “is Fake” (1 = fake, 0 = authentic)

X : variables prédictives (caractéristiques géométriques)

- **Split train test:**

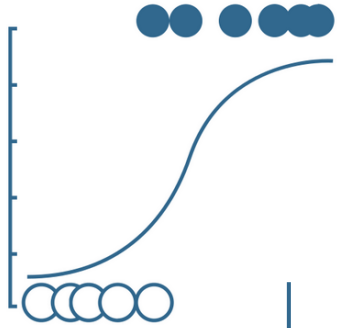


- **Normaliser les données:**

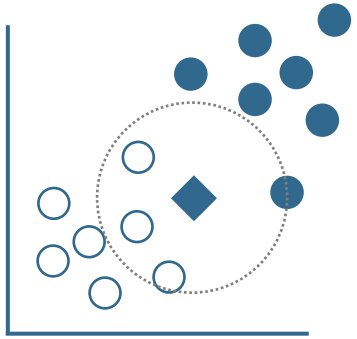


Les algorithmes de ML

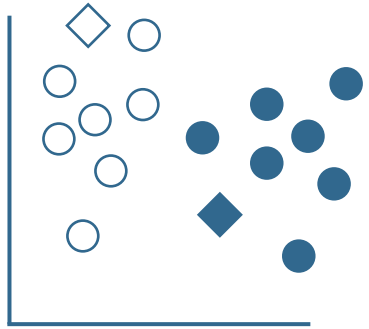
Logistic regression



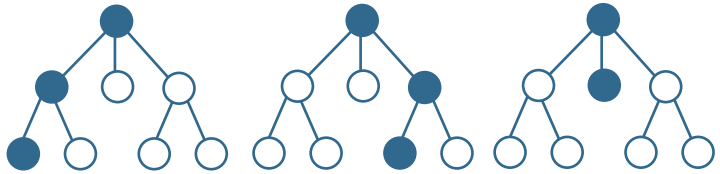
K-means



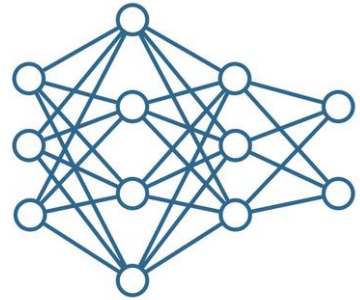
K-nearest neighbor (KNN)



Random Forest



Neural Network



Les algorithmes de ML

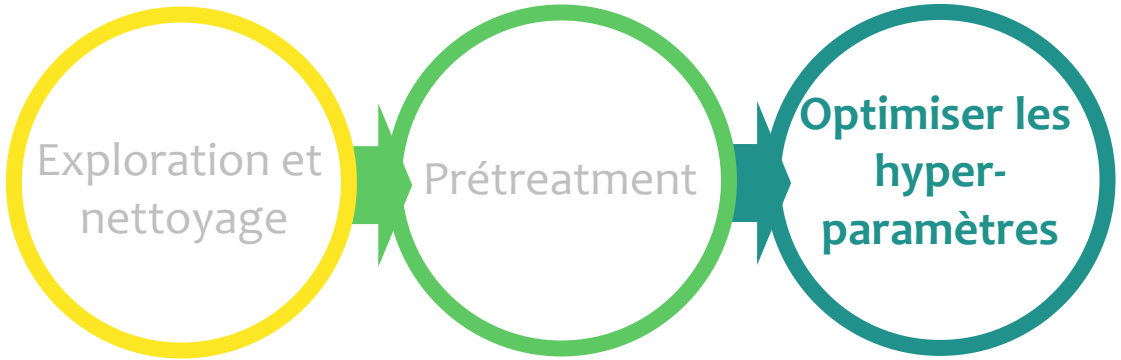
Logistic regression

K-means

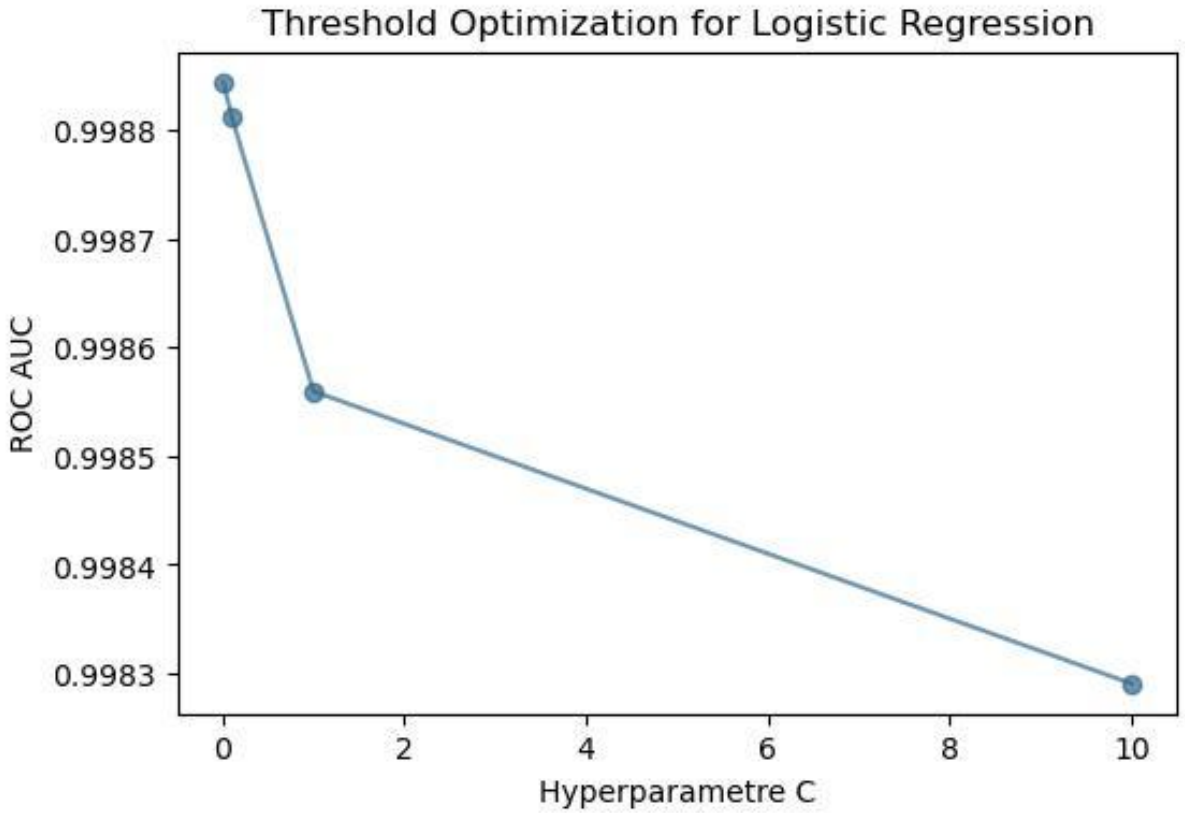
K-nearest neighbor (KNN)

Random Forest

Neural Network



LogisticRegression(C = 1, class_weight = “balanced”, etc…)



Les algorithmes de ML

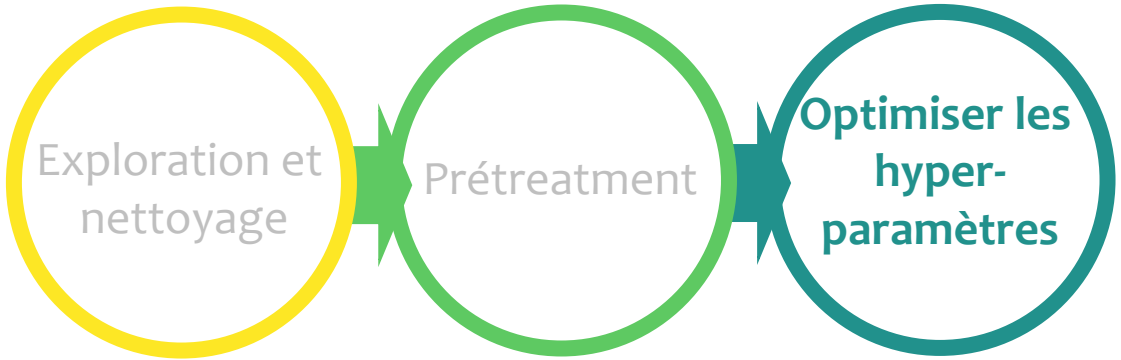
Logistic regression

K-means

K-nearest neighbor (KNN)

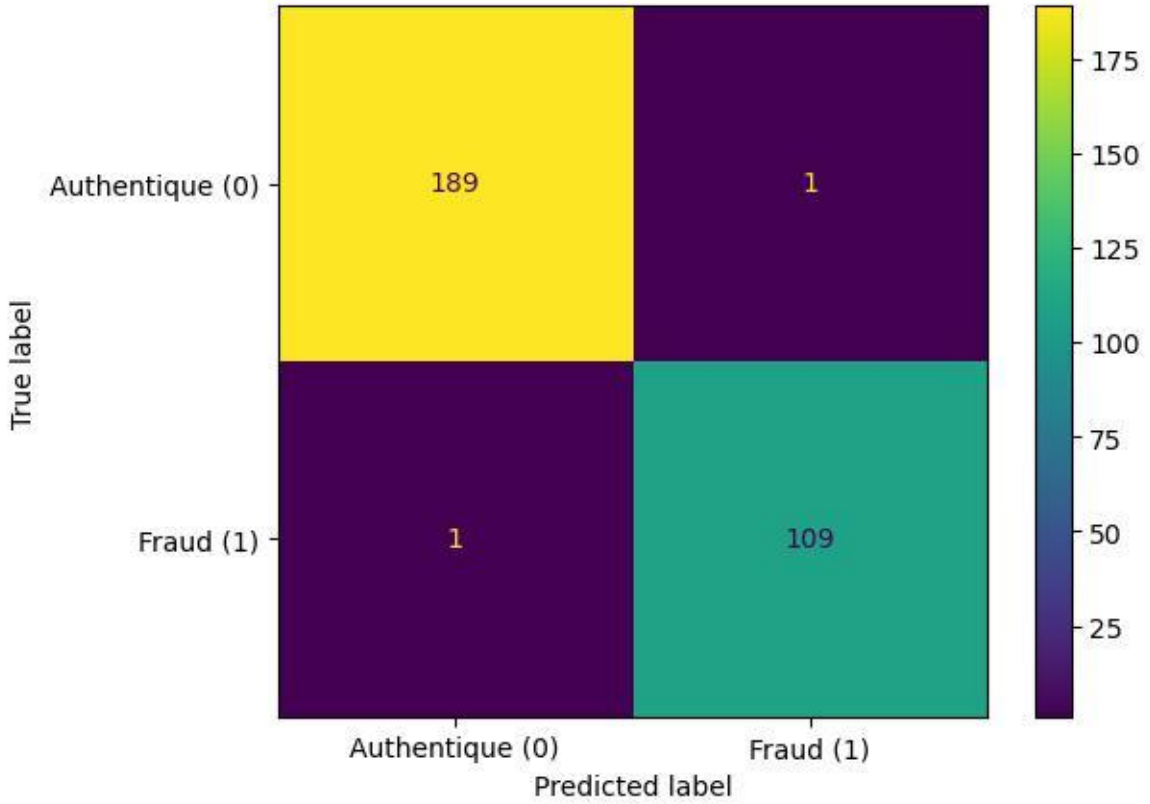
Random Forest

Neural Network



Comment savoir si le modèle est performant ?

Matrice de confusion



Accuracy = 0.99

Recall = 0.99

Precision = 0.99

F1 = 0.99

Les algorithmes de ML

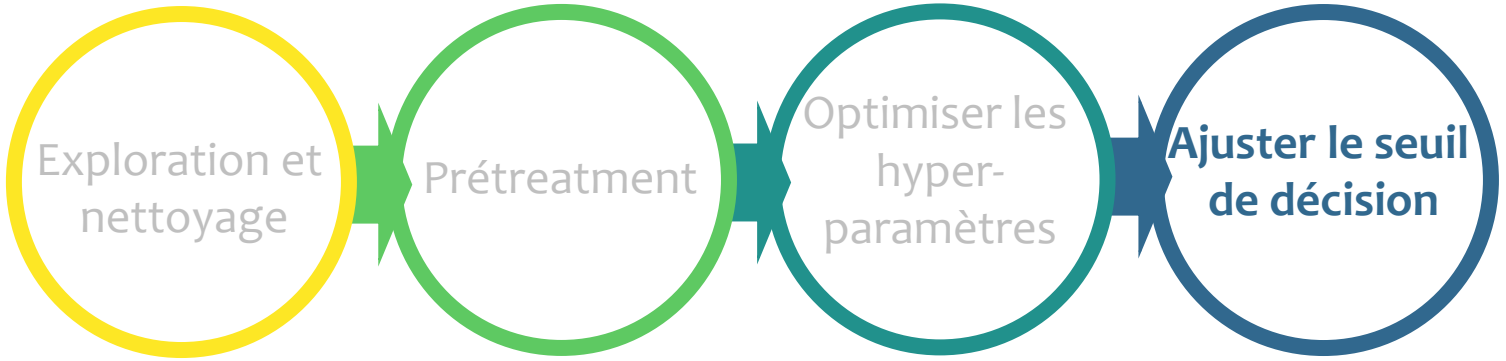
Logistic regression

K-means

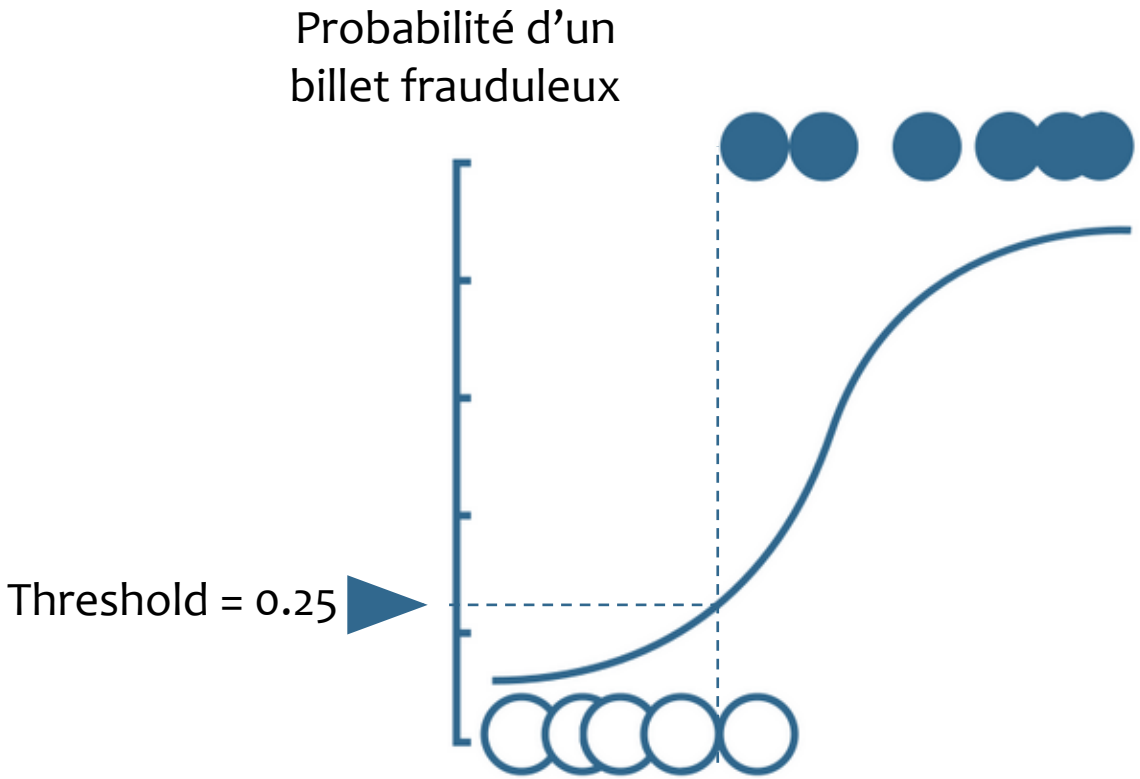
K-nearest neighbor (KNN)

Random Forest

Neural Network



Pour maximiser le recall



Les algorithmes de ML

Logistic regression

K-means

K-nearest neighbor (KNN)

Random Forest

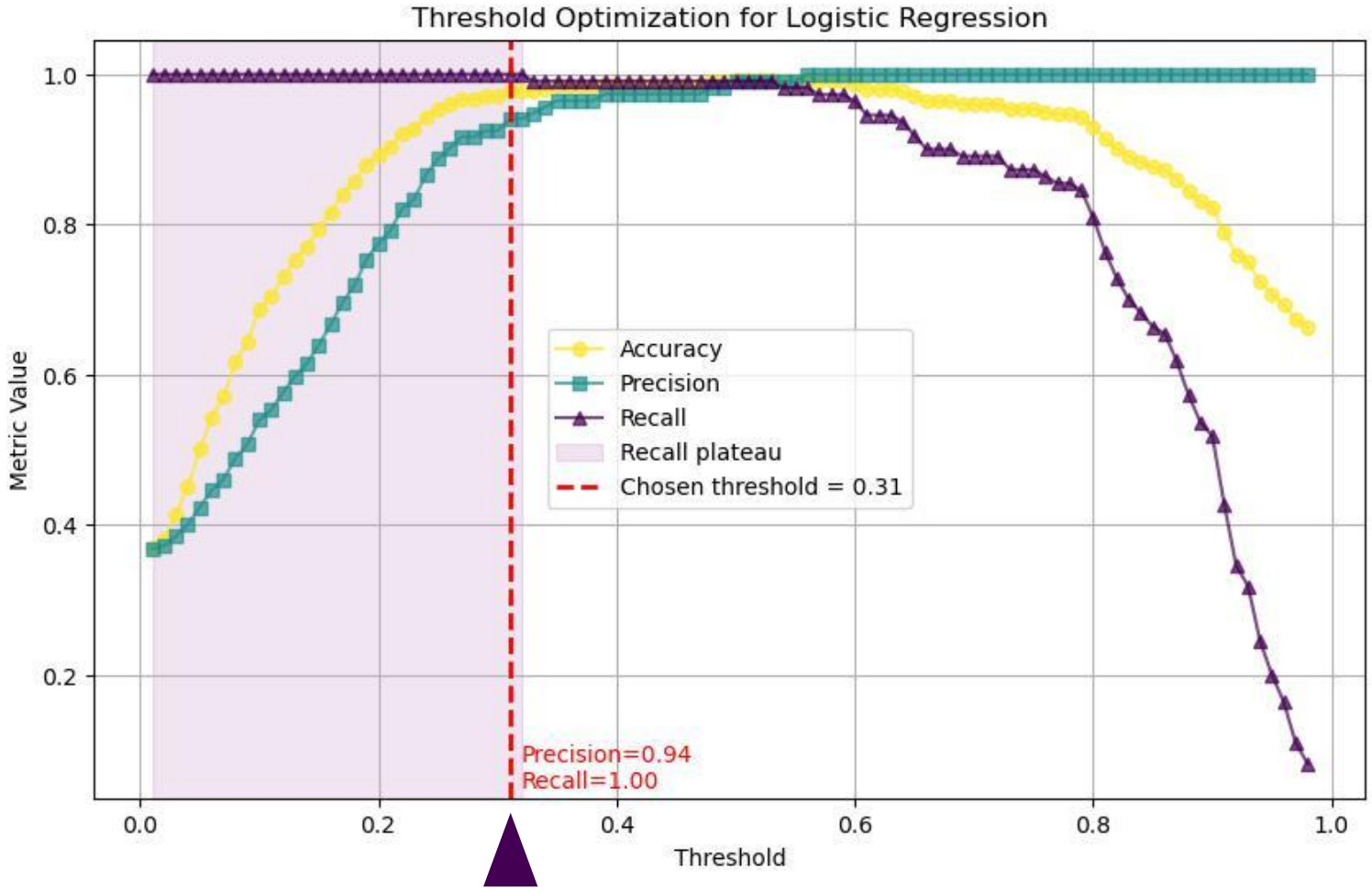
Neural Network

Exploration et nettoyage

Prétreatment

Optimiser les hyper-paramètres

Ajuster le seuil de décision



Les algorithmes de ML

Logistic regression

K-means

K-nearest neighbor (KNN)

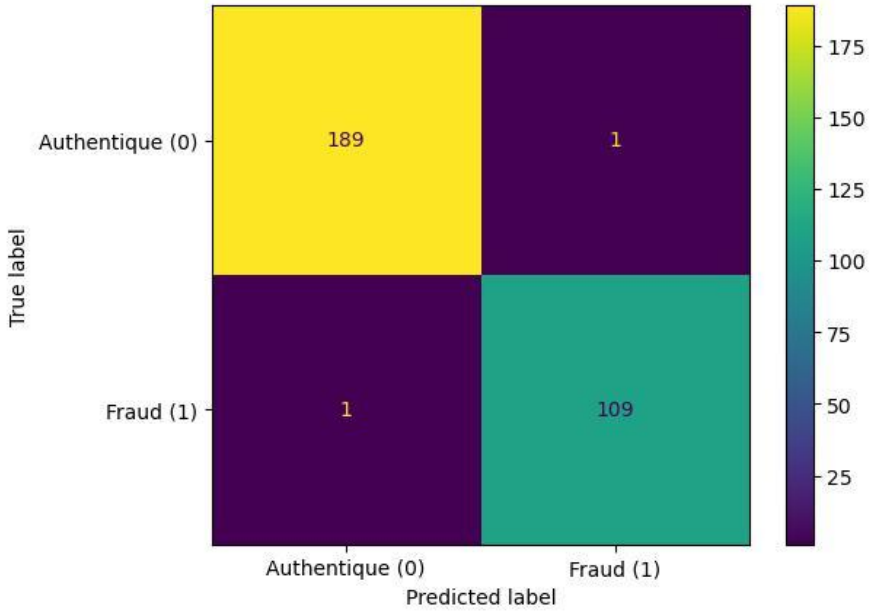
Random Forest

Neural Network

Exploration et nettoyage → Prétreatment → **Optimiser les hyper-paramètres** → **Ajuster le seuil de décision**

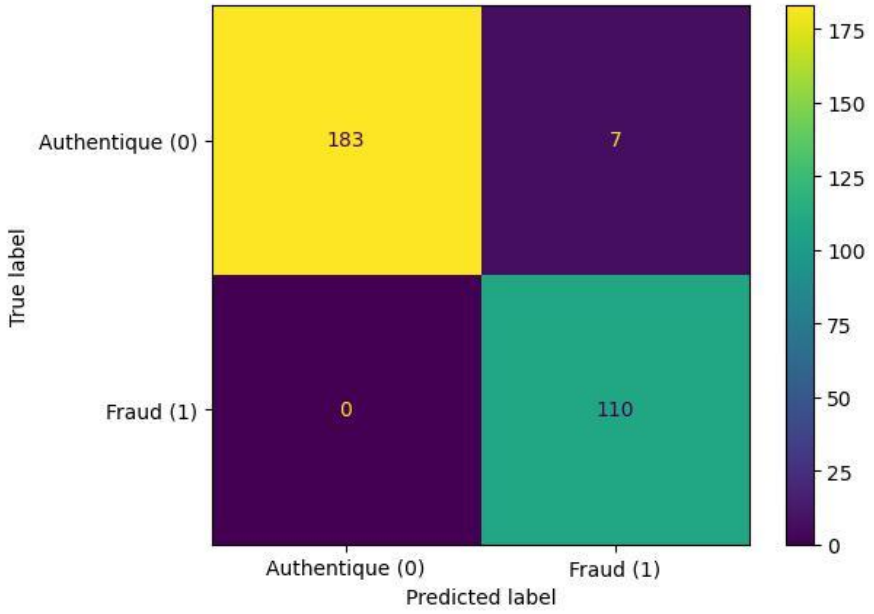
Matrices de confusion

Hyperparamètres optimisé



Accuracy = 0.99

Hyperparamètres et seuil optimisés



Accuracy = 0.98

Les algorithmes de ML

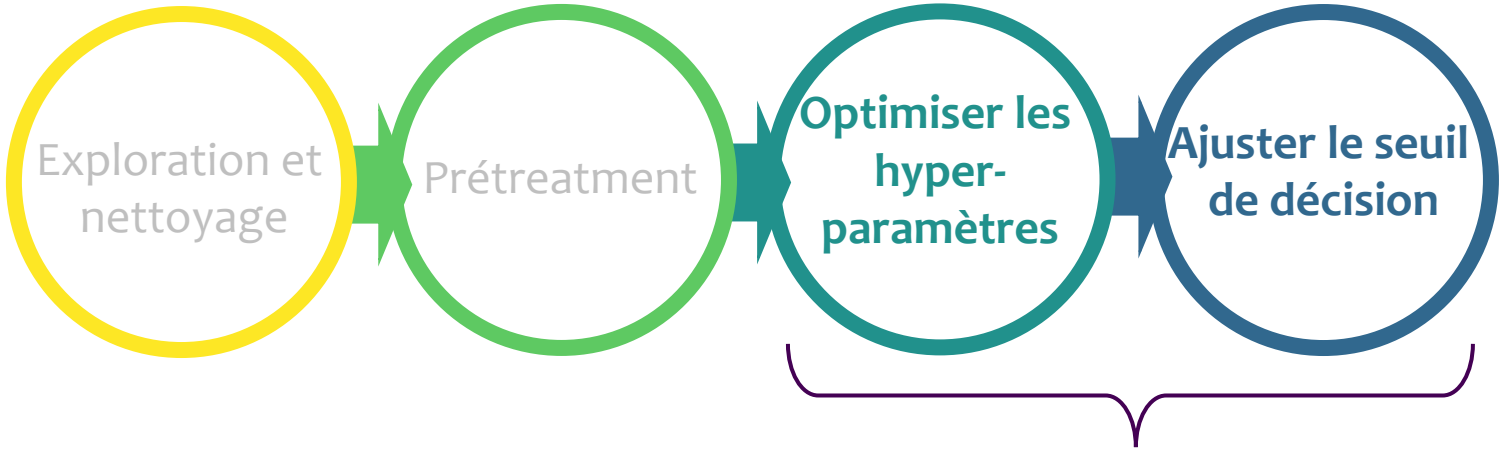
Logistic regression

K-means

K-nearest neighbor (KNN)

Random Forest

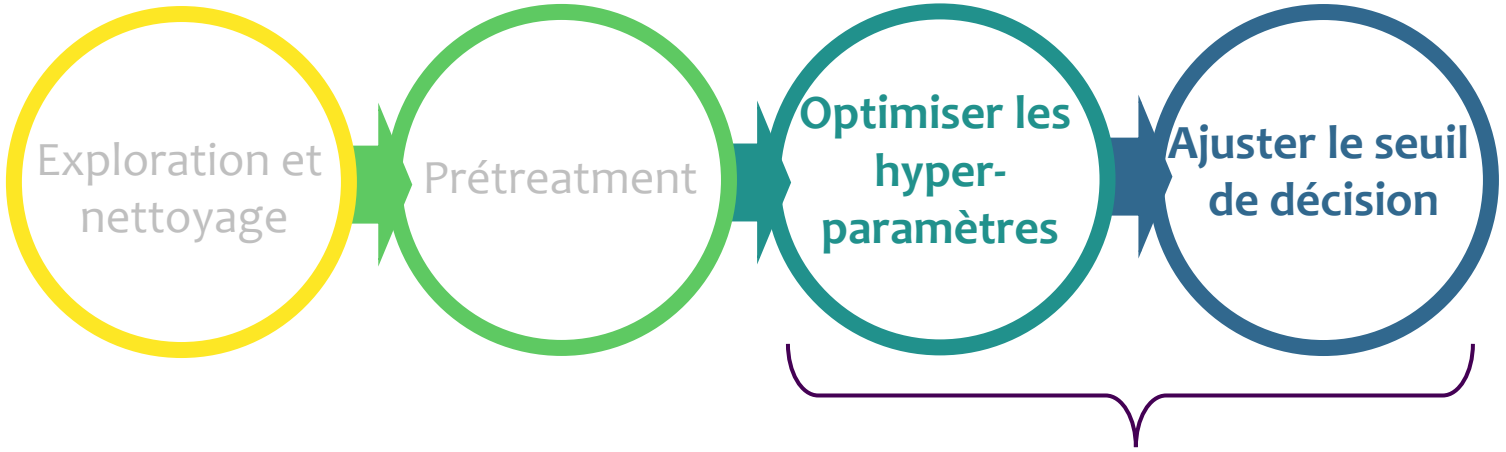
Neural Network



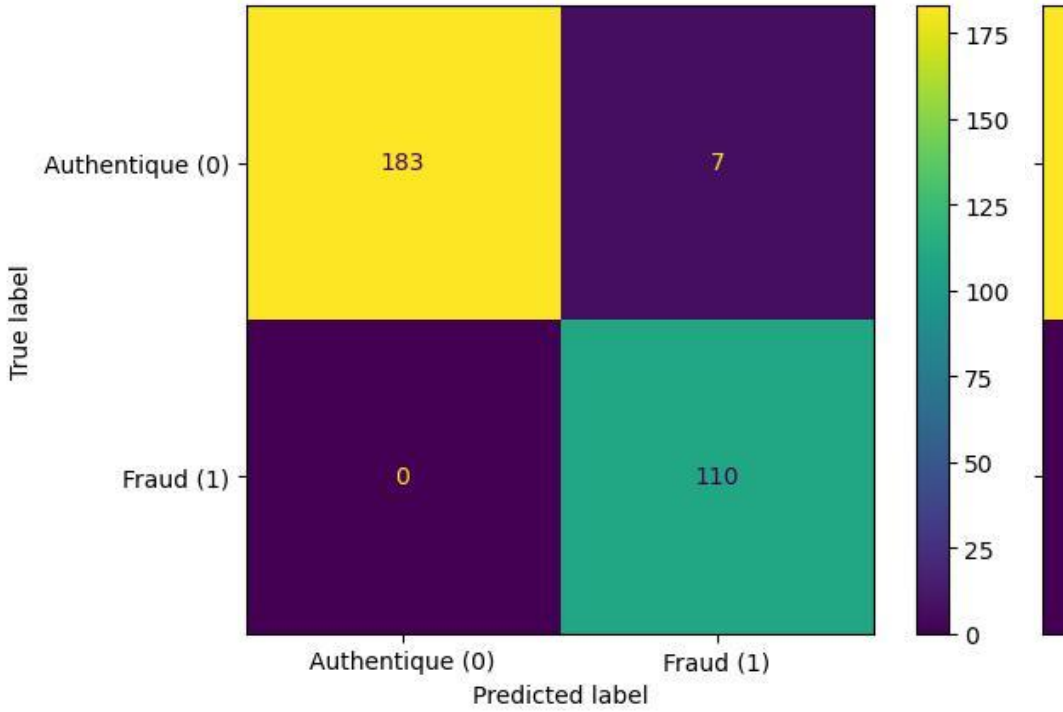
	AUC	Accuracy	Recall	Precision	F1
Regression logistique	0.9997	0.98	1.00	0.94	0.97
K-means		0.99	0.99	0.98	0.99
KNN	0.9998	0.99	1.00	0.97	0.99
Random Forest	0.9998	0.99	1.00	0.98	0.99
Neural Network	0.9998	0.98	1.00	0.95	0.97

Les algorithmes de ML

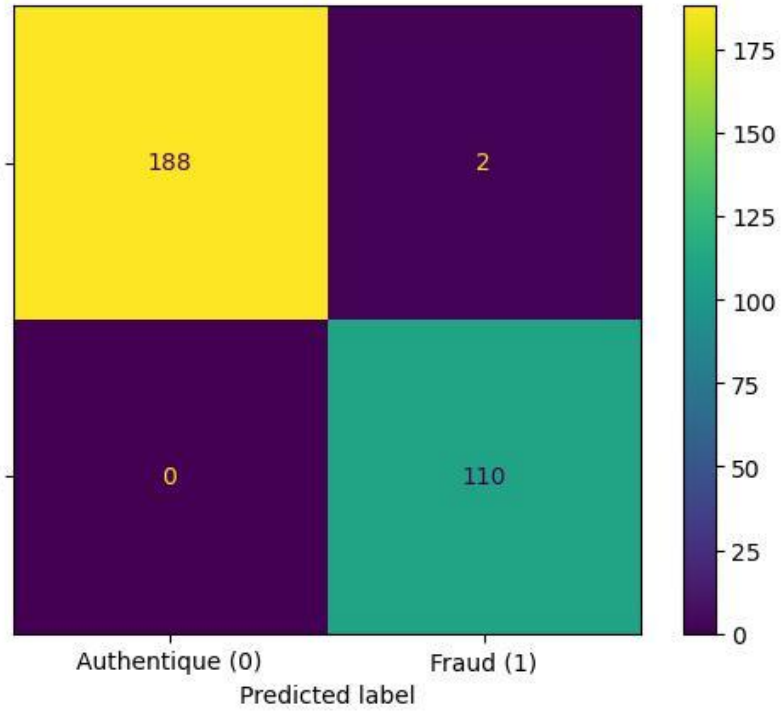
- Logistic regression
- K-means
- K-nearest neighbor (KNN)
- Random Forest
- Neural Network



Régression logistique



Random Forest



Les algorithmes de ML

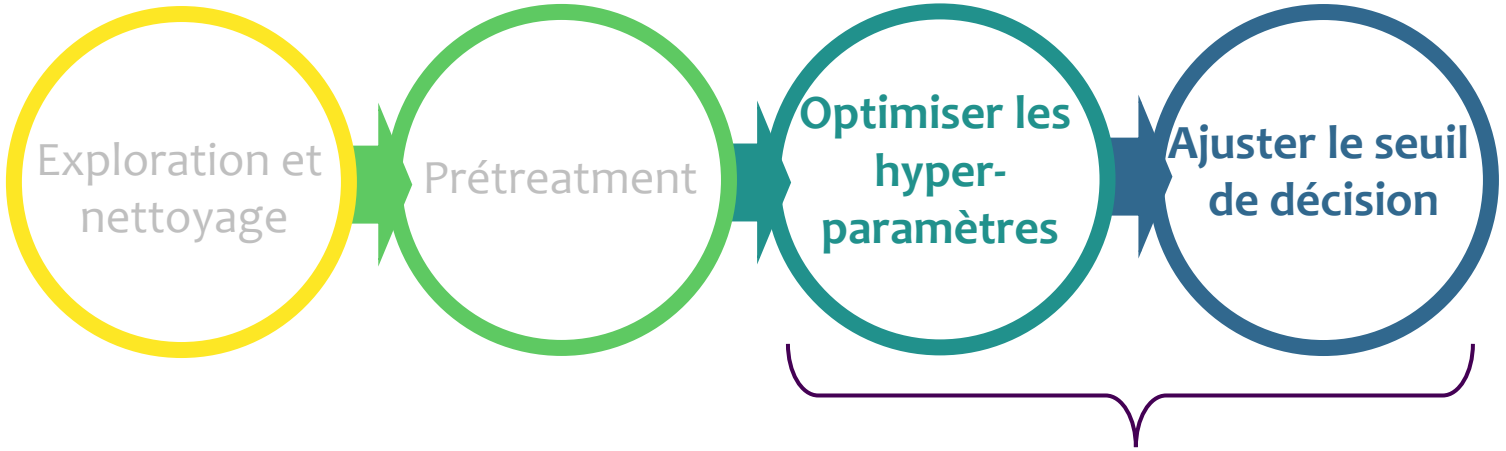
Logistic regression

K-means

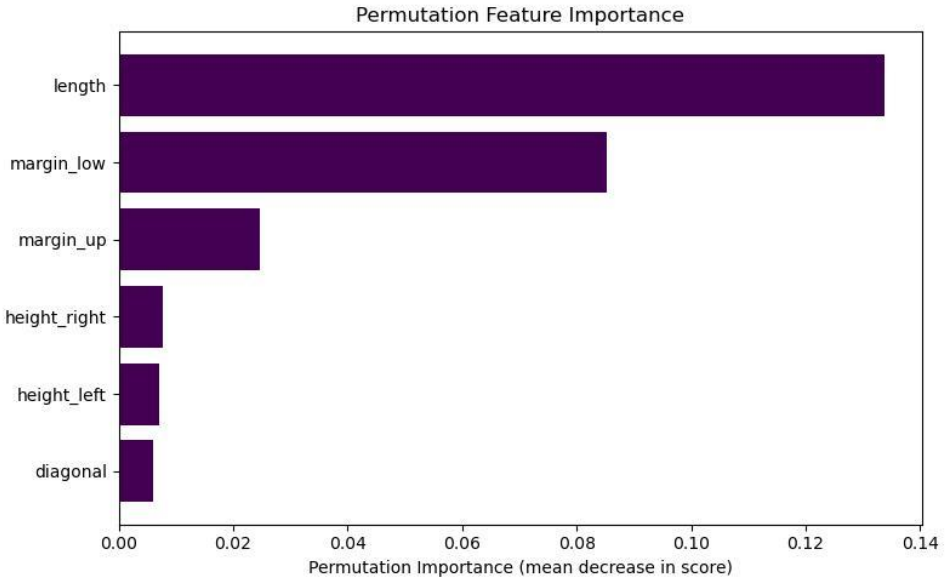
K-nearest neighbor (KNN)

Random Forest

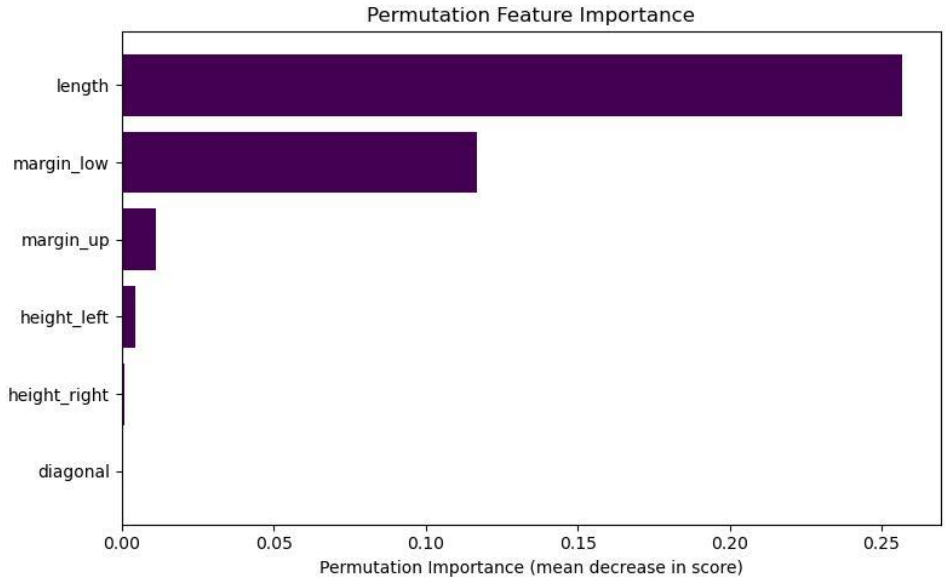
Neural Network



Régression logistique



Random Forest



Les algorithmes de ML

Logistic regression

K-means

K-nearest neighbor (KNN)

Random Forest

Neural Network



Modèle Machine Learning de Régression Logistique

Les algorithmes de ML

Logistic regression

K-means

K-nearest neighbor (KNN)

Random Forest

Neural Network

Conclusions

- Tous les modèles sont très **performants**
- L'algorithme des **K-means** est **moins adapté** à ce type de tâche (pas un classificateur, impossibilité d'ajuster le rappel et la précision)
- L'algorithme **Random Forest** a les meilleurs résultats. Recall parfait, et haute precision
- La **régression logistique** a été privilégiée en raison de **sa simplicité** (transparence, temps de calcul, etc). Recall parfait, mais precision légèrement inférieure
- Less is more!